



EEE423 INDUSTRIAL CONTROL

Mini-Greenhouse Automation Project



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1. INTRODUCTION

Greenhouse automation is an advanced technological system used in plant cultivation. With this system, environmental parameters such as temperature, humidity, light levels, and carbon dioxide concentration inside the greenhouse are monitored and controlled through sensors. Automation integrates irrigation systems, ventilation, lighting, and heating-cooling equipment to create optimal conditions for plant growth. By minimizing human intervention, it increases agricultural efficiency while saving energy and resources.

The growing global population limited natural resources, and the adverse effects of climate change on agriculture necessitate the adoption of modern technologies in farming. In greenhouse farming, it is critically important to continuously optimize the environmental conditions required for plant growth. While traditional methods fail to meet these needs, automation systems offer innovative and effective solutions for agricultural production. This project aims to develop an automation system that enhances efficiency in greenhouse farming and supports environmental sustainability. The developed system uses a control mechanism that integrates sensors and actuators to automatically regulate environmental factors in the greenhouse, ensuring ideal conditions for plant growth.

2. PERFORMED WORK

2.1 Hardware

The hardware block diagram of our project is presented in Figure 2.1.2, while the circuit schematic is shown in Figure 2.1.1. The power input section utilizes a 12VDC input voltage (for simplicity, a 12VDC 1A adapter was used) and is regulated using a 12V/12V&5V regulator circuit to supply various hardware components. While the sensors, which monitor environmental conditions, are explained in detail in the software section, the actuators include elements such as a fan, LED, and water pump. Additionally, buttons and switches are employed for control purposes.

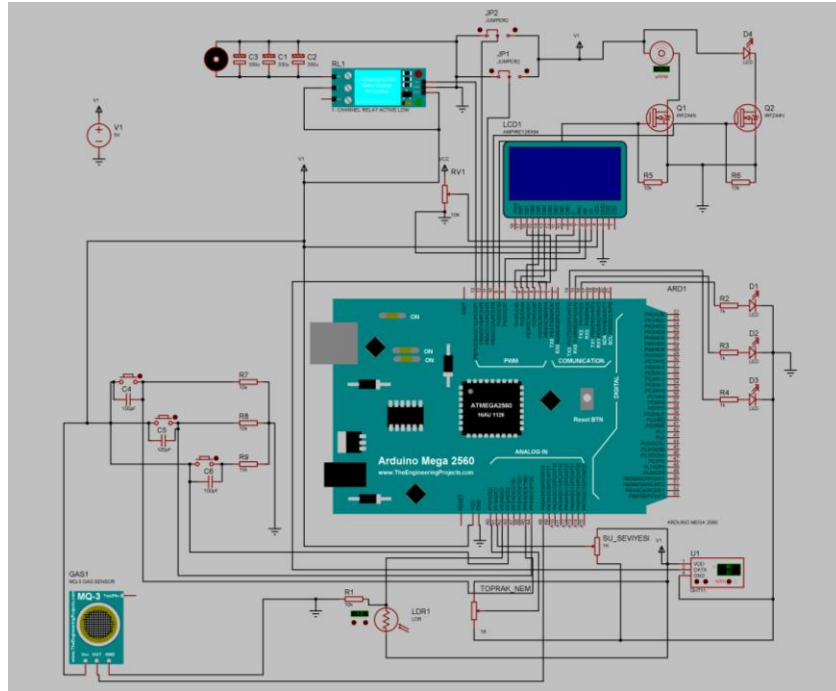


Figure 2.1.1 – Circuit Schematic

Furthermore, PWM amplifier MOSFET circuits and a water pump driver circuit are utilized. Due to the noise generated by the water pump in the Arduino system, the power supplies of the Arduino and the water pump were separated, and three $470\mu\text{F}$ capacitors were connected in parallel to the water pump terminals. For stability, buttons were supported with pull-down resistors and 100pF debounce capacitors. Lastly, the brightness of the LCD screen is controlled via a 10K potentiometer.

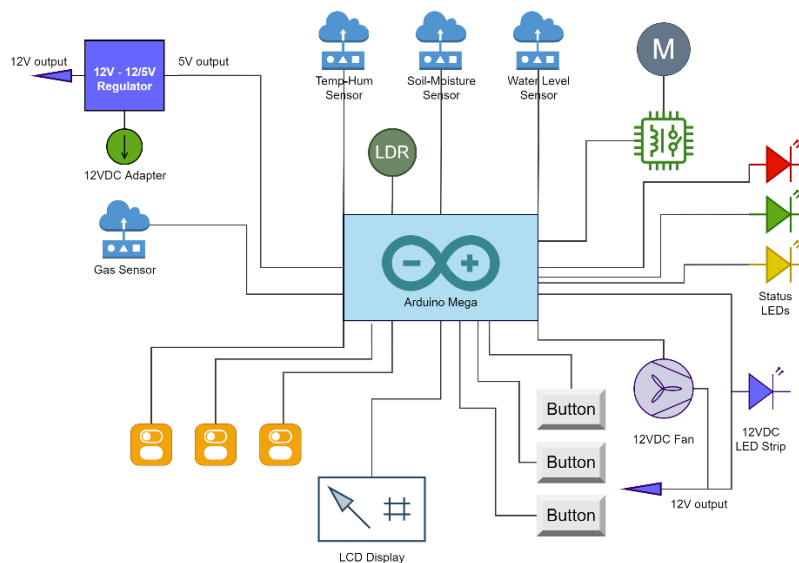


Figure 2.1.2 – Project Hardware Block Diagram

2.2 Software

The software in our project is designed to measure and automatically control environmental parameters such as temperature, humidity, soil moisture, water level, gas concentration, and light intensity. Additionally, it enhances user experience by offering manual user control and predefined scenarios (e.g., "Rainforest" or "Desert" modes). The software processes data from various sensors, monitors specific threshold values, and manages connected devices (fan, LED, irrigation motor). The microcontroller used in this system is an Arduino Mega. Sensor data are read and interpreted both analogically and digitally. The sensors employed in the system are as follows:

- DHT11: Measures temperature and humidity.
- MQ2 Gas Sensor: Monitors air quality.
- Soil Moisture Sensor: Detects moisture levels in the root zone of plants.
- LDR (Light Dependent Resistor): Measures ambient light intensity.
- Water Level Sensor: Monitors the water level in the tank.

The system controls the following components:

- Fan: Regulates temperature and air quality based on gas sensor readings.
- LED: Provides illumination depending on ambient light levels or operates according to user preferences.
- Irrigation Motor: Automatically waters the soil when the soil moisture level falls below a defined threshold.
- Status LEDs: Green, yellow, and red LEDs indicate the water level status.

The control of these components is implemented within the software through conditional statements and threshold values.

Our project offers users two primary control modes. The first is manual control, which provides options for adjusting LED brightness, fan speed, and threshold values via physical buttons. Additionally, it allows users to manage and configure these parameters through a user interface developed in C#. Figure 2.2.1 illustrates our bidirectional serial communication-based user interface. The second control mode consists of predefined modes: Desert Mode and Rainforest

Mode. For instance, the Rainforest Mode maintains low fan speed and high humidity, while the Desert Mode focuses on bright illumination and low humidity. These modes enable the cultivation of tropical or desert plants by tailoring environmental conditions accordingly.

Furthermore, the system continuously monitors the LDR (Light Dependent Resistor) values. When the switches activate the user mode, the lights automatically turn on in dark environments. Similarly, air quality is regularly assessed using the gas sensor, and if poor air quality is detected, the fan is activated automatically.

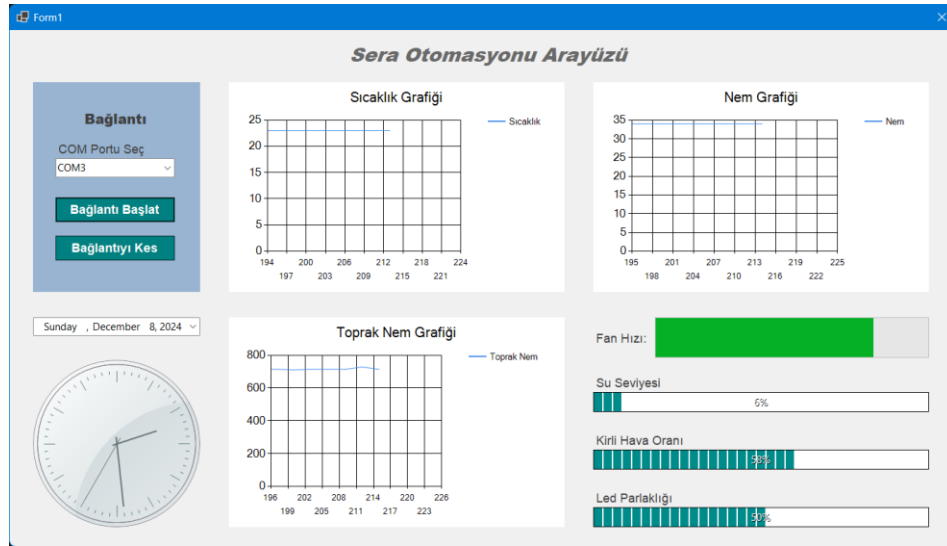


Figure 2.2.1 – Mini-Greenhouse Automation Project GUI

3. CONCLUSION

In our greenhouse automation project, we aimed to develop an innovative system that contributes to modern agricultural practices while providing both theoretical and practical experience. During the project, we encountered numerous hardware and software challenges but addressed them with a systematic approach. For example, although we connected all components according to the circuit diagram and linked them to the Arduino, we noticed that the relay was continuously switching, the screen was not displaying the expected data, and the buttons were unresponsive. Suspecting that the issue might be caused by electrical noise, we added two capacitors in parallel to the motor and eliminated the noise problem. To address potential difficulties with the buttons, we designed a greenhouse automation interface that allowed control through the interface. After

further iterations, we refined the code and observed that everything was functioning as intended. This project has not only enhanced our technical skills but also provided opportunities to develop our problem-solving, teamwork, and innovative thinking abilities. From integrating sensors and actuators to developing software algorithms, we gained valuable experience in designing effective and efficient systems for real-world applications. Additionally, we deepened our understanding of environmental sustainability, learning how to achieve greater yields with fewer resources. This project has demonstrated the potential of greenhouse automation to transform agricultural practices while fostering a more sustainable and efficient future for farming.

4. EXTRAS

Arduino Codes:

https://drive.google.com/file/d/10_PN3F_tvHe-Fh8Fg-oTJgiHatgQJ9Zr/view?usp=sharing

C# GUI (Graphical User Interface) Codes:

https://drive.google.com/file/d/1WDVc7X_O36V1JTogVINDG9C6y8Tz3akI/view?usp=sharing

A Video of the Project While Working:

https://drive.google.com/file/d/181UUulZH5IvQYiWF5yPLu6etpXgM_r7w/view?usp=sharing